Design an Accurate BCI System by a Novel Method and the Minimum Number of Channels Based on Five Class Motor Imagery

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***Abstract*—Motor imagery (MI) is one of the most popular methods to control a BCI system. Each specific MI can be considered as an individual task and a command for controlling an external device. Common spatial pattern (CSP) is one of the most popular methods for feature extraction of MI. This method filters EEG in spatial space and it can be useful for MI recognition. In this project, EEG was preprocessed by common average reference (CAR), bandpass Butterworth and bandstop filter were used for removing artifacts and noises. For feature extraction, filter bank common spatial pattern was used. Sequential forward feature selection method was used to select best features and also best channels. After feature selection, difference between features was used and then linear discriminant analysis, K nearest neighborhood and support vector machine were used for classification. BCI-Competition iv and also new data were collected were used in this project. The accuracy for new data was 86.72% mean for 25 people and 99.00% mean for 9 subjects in BCI-Competition iv.**

Keywords— Classification, Difference between Features, Feature Slection, Filter Bank Common Spatial Filter, Motor Imagery

I. Introduction

A brain-computer interface (BCI) is a system that takes and translates brain signals into commands for a device. These systems can help paralyzed and disabled people but this isn't the only usage of BCI. BCIs can be more useable in nowadays world for increasing the speed of works with only think about actions and not to operate. BCI systems are generally based on three types and the combination of these types. BCIs based on MI, BCIs based on steady state visual evoked potential (SSVEP) and BCIs based on P300 signals. MI has an advantage in compare with the other methods that is it doesn't need external simulations but generally the accuracies in BCIs based on MI are less than other types.

The human nervous system consists of two main subsets; The central and the peripheral nervous system. Brain is the most important part of central nervous system and it controls the human body with electrical signals that propagates throw the peripheral nervous system. BCI provides a direct communication between a computer and human brain and so it can act like a peripheral nervous system and the human body. Therefore, a BCI system could help paralyzed people control a wheelchair or a car and this is just an example of BCI.

BCI systems are formed of two main parts. First part is a system that it collects brain's data and processes that and the second part is a system that performs the commands. One of the most important brain signals is EEG because it is low-cost and also non-invasive. EEG is nonstationary because of changes of electrodes features such as impedance or positions, subject's attention, fatigue, eye movements, or muscular activity, EEG signals exhibit large subject variation [1,2].

Motor imagery is one of the most popular methods to control BCI systems. MI is the act of imagining a specific action without executing it [3]. Temporal, spatial and frequency features are useful to classify MI. Motor imagery events have been observed in the somatosensory cortices and in mu (8-13) and beta (13-30) rhythm [4,5]. The regions that active during MI are shown in Fig. 1. They are placed mostly in the center of the brain cortex and central sulcus [3]. This is important to use appropriate electrodes for extracting MI signals. Signals which are generated from MI are similar to the execution of individual motor actions. It can be said that MI is a good and easy way to generate individual signals for controlling a device. One of the most important challenges in designing BCI systems based on MI is low accuracy and also the number of channels. A good BCI system is a system that is the most accurate and fastest system with the minimum number of channels.

In this research, a wrapper method that is sequential feature selection toward forward (SFFS) method was used to select best channels so the minimum number of channels can be used for recording MI data. This method is a wrapper method for feature selection. Difference between features (DBF) method is a novel method that is proposed in this article to increase accuracy for MI classification. In this method the difference between features are used instead of the features themselves.

The next part of this article is about related works in section II and then the material and methods are described in section III. In section III, the DBF method is described. In section IV, the results are shown and discussed and the results without and with using DBF are compared. In section V, the results are concluded.

II. Related Works

Common spatial filter and other methods like wavelet transform (WT), short time Fourier transform (STFT) and auto regressive (AR) models were used for feature extraction from motor imagery. WT was used on BCI-competition number two

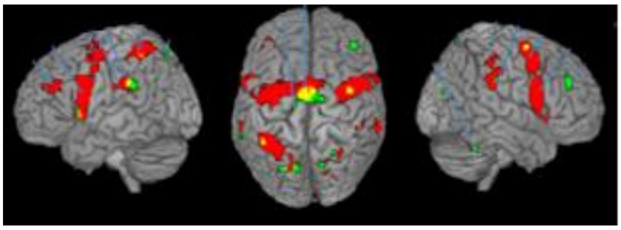

Fig. 1. The sensory motor regions in the human's brain [3]

and BCI-competition number three data and accuracies were 90% [6]. In this paper, the neural network which is convolutional (CNN) that is a complicated method, used for classification. This method computationally is expensive and it takes long time to process.

A lot of papers can be found that used CSP for feature extraction. The accuracies were 78.55% for BCI-competition number four data and 86.6% for BCI-competition number three data [7]. The kappa was 0.627 for BCI-competition number four. In this paper, support vector machine (SVM) was used for classification.

A lot of algorithms based on CSP like adaptive CSP (ACSP) [8] are used for motor imagery classification and the kappa for BCI-competition number four was 0.52. Time frequency CSP (TFCSP) was used for feature extraction of MI data and the kappa for BCI-competition number four was 0.52 [5].

The method that based on source localization [9] obtained good accuracies were achieved from BCI-competition iv. The accuracies are about 85%. In this paper, visibility graph (VG), Kalman filter and source localization methods are used. Although this paper achieved good accuracies, but the method is very complicated and the computation cost is a lot. Time is an issue in BCI systems and we should consider the processing time for practical systems.

The past projects mostly used BCI-competition data for designing BCI system based on MI. In this project, BCI-competition iv 2a and also the recorded data in Amirkabir University of Technology were used.

III. MATERIAL AND METHOD

In this research, common average reference (CAR) filter and a butterworth bandpass filter and a bandstop filter for removing noises and artifacts were used. After preprocess, filter bank common spatial pattern (FBCSP) applied to the signals and the features were extracted and selected with sequential forward feature selection (SFFS) method. After feature selection, the differences between features calculated as the DBF and for classification, linear discriminant analysis (LDA), k-nearest neighborhood (KNN) and support vector machine (SVM) were used. In Fig. 2, block diagram of this research is shown.

A. Datasets

BCI-competition iv 2a [10] is a very famous dataset for motor imagery tasks. This is a four class data and consists of imagination of left hand, right hand, both foot and tongue. In this dataset, motor imagery trials were recorded from 9 subjects in two sessions. The recording protocol is shown in Fig.3. All information about this dataset can be reached in reference.

Another dataset was used in this project recorded from 25 people in Amirkabir University of Technology with mean and standard deviation age 25.28+-1.18 year. This is a fifth class recording and the classes were left hand motor imagery, right hand motor imagery, both legs motor imagery, tongue motor imagery and teeth clenching. Each person imagined 60 trials for each class so the total number of trials is equal to 300. Each trial consists of five steps; step one is an Announcement for the imagery, step two is a 300 milliseconds delay for subject reaction delay, step three is a beep voice that is announced the beginning of recording, step four is one second recording EEG data and the step five is an another 300 milliseconds delay for a break. The recording protocol is described in Fig. 4. The electrodes were F3, Fz, F4, Fc3, Fcz, Fc4, C3, Cz, C4, Cp3, Cpz, Cp4, P3, Pz, P4, T7, T8, O1, Oz, O2. The sample frequency was 250 Hz.

B. Preprocessing

Common average reference filter was used to remove common activities and left the individual activity of each EEG channel [11]. After implementing CAR filter, output is described by Eq. (1), where $X_{i(t)}^{CAR}$ is the ith channel filtered output, $x_i(t)$ is ith channel and c is the total number of channels.

$$X_{i(t)}^{CAR} = x_i(t) - \frac{1}{c}\sum_{j=1}^{c} x_j(t) \qquad (1)$$

the mean activity was removed from each channel. Then a 3^{rd} degree butterworth bandpass filter that it's bandwidth is between 4 to 40 Hz was used, then a bandstop 3^{rd} degree with 49-51 Hz cutoff frequency was used for removing line noise. The preprocessing stages are shown in Fig. 5. A one second part of the C4 channel's signal before and after preprocessing is shown in Fig. 6.

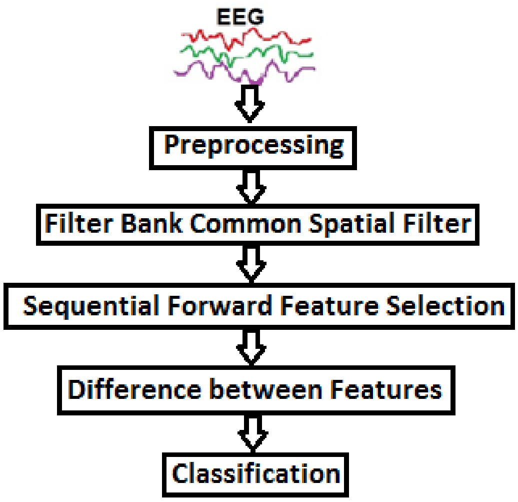

Fig. 2. Block diagram of this research

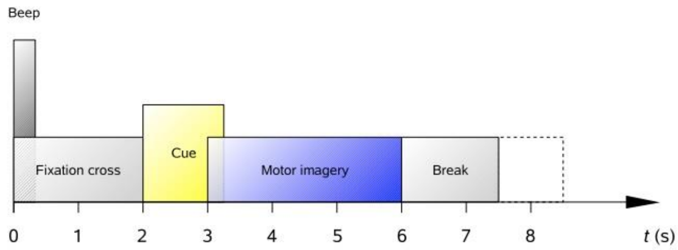

Fig. 3. Recording protocol in BCI-competition iv

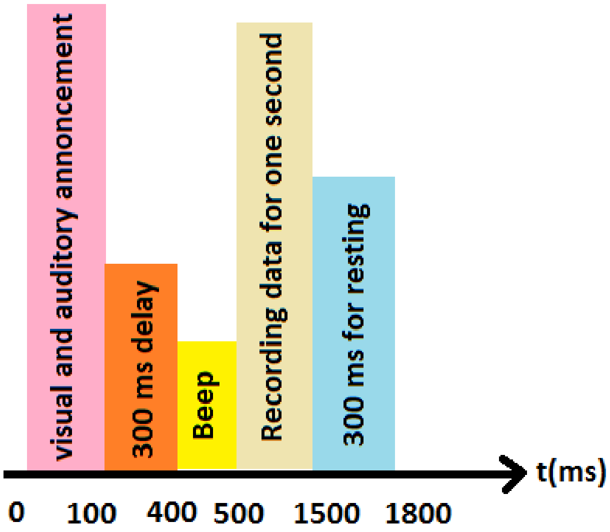

Fig. 4. Recording protocol in this research

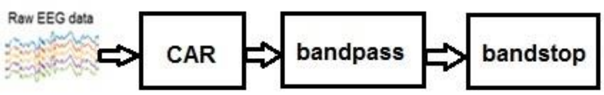

Fig. 5. Preprocessing stages

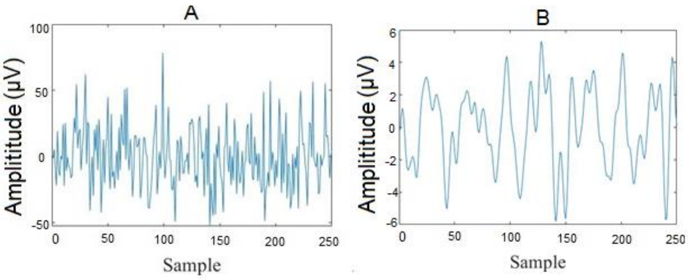

Fig. 6. A. C4 channel before preprocessing, B. C4 channel after preprocessing

C. FBCSP

CSP is a spatial filtering technique that is popular and useful for MI recognition. Originally, CSP is a two-class method that transfers high dimensional EEG data to a low dimension space aims to maximize the feature variation of one class while minimizing the feature variation of the other class [2]. With CSP, a linear transform is used to project the multi-channel EEG into a lower dimension with a projection matrix [12].

The details are described as follows. consider x_l and x_r as left-hand and right-hand trial matrix with dimension $N \times T$ where the quantity of channels is equal to N is and T is the number of samples which each channel contains. The spatial covariance of the EEG which is normalized can be represented as follows:

$$R_r = \frac{x_r x_r^T}{\text{trace } (x_r x_r^T)} \tag{2}$$

$$R_l = \frac{x_l x_l^T}{\text{trace } (x_l x_l^T)} \tag{3}$$

These equations show that how it is possible to calculate the normalized spatial covariance. x^T is the transpose of x and trace(A) calculates the sum of the diagonal elements of A. We should compute the mean of normalized covariance $\overline{R_r}$ and $\overline{R_l}$ by averaging over all trials of each group. Composite spatial covariance can be factorized as follows:

$$R = \overline{R}_r + \overline{R}_l = U_0 \Sigma U_0^T \tag{4}$$

where U_0 is the matrix of eigenvectors and Σ is the diagonal matrix of eigenvalues. Therefor we can define P as the whitening transformation matrix.

$$P = \Sigma^{-\frac{1}{2}} U_0^T \tag{5}$$

The whitening transformation matrix can be used as follows:

$$S_r = P\overline{R_r}P^T \tag{6}$$

$$S_l = P\overline{R_l}P^T \tag{7}$$

The eigenvectors with the smallest eigenvalues for S_l have the largest eigenvalues for S_r and vice versa. The projection matrix W is denoted as follows:

$$W = U^T P \tag{8}$$

Original EEG can transform into uncorrelated components with W as follows:

$$Z = WX \tag{9}$$

EEG source components are placed in Z. The rows of W matrix in CSP algorithm, consist of spatial filters that the first and the last rows are the most important CSP filters; Because they represent the largest variance of one task and the smallest variance of the other task. However, the second, third, and fourth rows from the top and below of the W matrix were used to use more features for classification. The number of selected rows is a trade-off between accuracy and calculation complexity. Common spatial pattern can be used as a multiclass algorithm by using one versus others structure. So it can be transpose four class MI tasks to a new space in order to maximize feature variance in a class and minimize it in other classes. After applying CSP to the signal, logarithmic function was used to make the distribution like Gaussian.

Filter bank common spatial pattern is a filter bank that applies to the signal and after that CSP applies to the signal. In this project, 9 bandpass filter with 4Hz bandwith were considered. So the bands are 4-8Hz, 8-12Hz, 12-16Hz, 16-20Hz, 20-24Hz, 24-28Hz, 28-32Hz, 32-36Hz and 36-40Hz. we applied CSP on each band then extracted features.

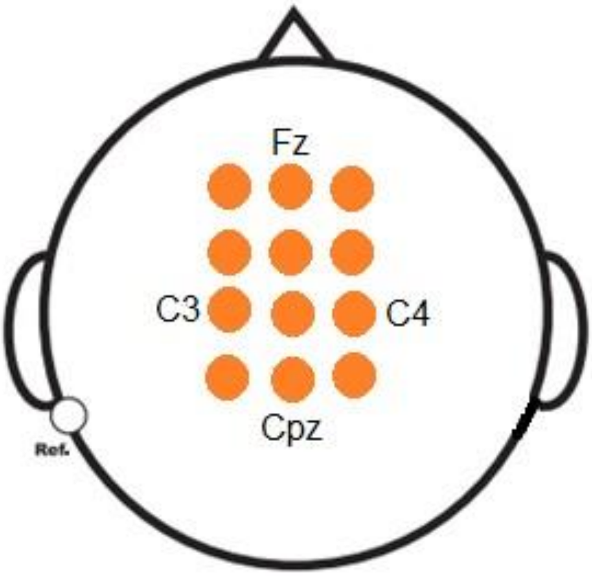

Fig. 7. Selected channels with SFFS method

D. Channel and Feature Selection

After feature extraction, SFFS was used for selecting best channels and also was used to find best features. Channel Cz was considered as the reference and after that other channels were considered with Cz to find best channels. Each channel was considered beside Cz to find best combination of channels which conclude more accuracy for classification with LDA classifier. This is the basic of SFFS method that it must be considered one classifier and test all combination of elements to conclude more accuracy for classification. The best channels were F3, Fz, F4, Fc3, Fcz, Fc4, C3, Cz, C4, Cp3, Cpz and Cp4. Chosen electrodes are described in Fig. 7. After applying SFFS to the features, we chose 35 best features for classification.

E. Difference between Features

This method is usable to calculate the differences between features. In this method, one of the classes has to be considered as the reference class; in this method, if the reference class is more separable, the results will be better. In this research teeth clenching for recorded data is considered as the reference class and imagination of both foot movement is considered as the reference for BCI-competition iv data. After reference selection, 10 trials of these classes were extracted and all trials of all classes mines from these 10 trials and the mean of the results were calculated and now these amounts are considered as the features. In other words, in this method we use differences between features and the reference class features instead of the exact features.

F. Classification

The features were extracted from MI data are useful for classifying the MI tasks. LDA in spite of being a simple classifier, is an effective classifier and it's comparable to other nonlinear and more complicated classifiers such as SVM. LDA, KNN and SVM have been used in this project. Hold-Out method was used for this classification and all subjects were considered together. 70% data were used for training and 30% data were used for test. Euclidean distance was used for all classifiers and the kernels for SVM were linear, quadratic, cubic, fine Gaussian, medium Gaussian and coarse Gaussian and the kernels were used for KNN were fine, medium, coarse, cosine, cubic and weighted. The number of neighbors for KNN was equal to 5 for all experiments.

IV. RESULTS

Kappa and accuracy are described below show that a BCI system can be designed with the minimum number of channels. In proposed method a BCI system was designed with only 12 channels that classify a four and a five class problem with good accuracy. To compare with other works, a five class system is more usable for controlling aspects. The most important innovation in this research is the using of the DBF method. The results for classification without using DBF on BCI-competition iv are described in TABLE I and with using DBF for this data are described in TABLE II. The same results for recorded data are described in TABLE III and TABLE IV. Results show that DBF is very useful for increasing accuracy. With DBF, the difference between trials are eliminated but the similarities are remained and this is useful for a better classification.

Kappa has been calculated as follows:

$$\text{kappa} = \frac{p_0 - p_e}{1 - p_e} \tag{10}$$

which p_0 is equal to number of agreement divided by the total number and p_e is equal to sum of the correct and incorrect. Accuracy can be calculated as follows:

$$\text{accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{11}$$

which the number of true positive is equal to TP recognition, T_N is the number of true negative recognition and the total is the total number of trials.

The results for classification for recorded data with using DBF method are described in TABLE II. With DBF method for this data, mean of accuracy has been increased about 10%. With DBF method, the mean of accuracy for BCI-competition data has increased about 50%. The mean of accuracy with this method is 99% and the kappa is 0.97 that describes a good method.

V. CONCUSION

The results show that DBF method increases the accuracy about 10% for all classifiers for recorded dataset and increases the accuracy about 50% for BCI-Competition iv data. The best classifier for both method is quadratic SVM that obtained 86.72% for DBF method and 75.68% without using DBF method on recorded data and obtained 57.31% and 100% for BCI-competition iv. Also with SFFS method for feature selection, it has been proved that central electrodes such as C3, Cz and C4 are more important than the peripheral electrodes like P3, Pz, P4 or T7 and T8.

TABLE I. ACCURACY WITHOUT DBF ON BCI-COMPETITION DATA

classifier	Accuracy (%) (Mean±STD)	Kappa (%) (Mean±STD)
LDA	54.25±3.69	0.52±0.06
Linear SVM	55.22±3.65	0.52±0.07
QuadraticSVM	**57.31±3.56**	**0.55±0.06**
Cubic SVM	52.78±3.73	0.50±0.08
FineGaussian SVM	49.12±3.74	0.48±0.06
Medium GaussianSVM	50.27±3.71	0.49±0.07
Coarse GaussianSVM	51.87±3.63	0.50±0.07
Fine KNN	46.20±3.66	0.44±0.07
Medium KNN	45.26±3.68	0.42±0.06
Coarse KNN	46.68±3.72	0.43±0.06
Cosine KNN	41.64±3.78	0.39±0.08
Cubic KNN	42.80±3.60	0.40±0.07
WeightedKNN	49.94±3.37	0.47±0.06
Mean	**49.48**	**0.47**

TABLE II. ACCURACY WITH DBF ON BCI-COMPETITION DATA

classifier	Accuracy (%) (Mean±STD)	Kappa (%) (Mean±STD)
LDA	**100.00±0.00**	**1±0.00**
Linear SVM	**100.00±0.00**	**1±0.00**
Quadratic SVM	**100.00±0.00**	**1±0.00**
Cubic SVM	**100.00±0.00**	**1±0.00**
Fine Gaussian SVM	**100.00±0.00**	**1±0.00**
Medium Gaussian SVM	**100.00±0.00**	**1±0.00**
Coarse Gaussian SVM	**100.00±0.00**	**1±0.00**
Fine KNN	98.45±0.58	0.94±0.03
Medium KNN	96.72±1.12	0.92±0.05
Coarse KNN	97.38±0.84	0.93±0.04
Cosine KNN	95.62±0.98	0.91±0.05
Cubic KNN	98.89±0.68	0.97±0.02
Weighted KNN	**100.00±0.00**	**1±0.00**
Mean	**99.00**	**0.97**

TABLE III. ACCURACY WITHOUT DBF ON RECORDED DATA

classifier	Accuracy (%) (Mean±STD)	Kappa (%) (Mean±STD)
LDA	70.34±3.07	0.69±0.05
Linear SVM	71.82±2.87	0.70±0.04
Quadratic SVM	**75.68±2.93**	**0.73±0.04**
Cubic SVM	74.49±2.80	0.73±0.04
Fine Gaussian SVM	72.53±3.12	0.71±0.05
Medium Gaussian SVM	73.86±2.92	0.72±0.05
Coarse Gaussian SVM	75.12±3.26	0.73±0.06
Fine KNN	68.90±2.75	0.64±0.04
Medium KNN	60.21±3.38	0.61±0.07
Coarse KNN	58.34±3.62	0.56±0.07
Cosine KNN	58.62±3.38	0.56±0.06
Cubic KNN	62.59±3.60	0.59±0.07
Weighted KNN	74.77±3.01	0.70±0.05
Mean	**69.02**	**0.66**

TABLE IV. ACCURACY WITH DBF METHOD ON RECORDED DATA

classifier	Accuracy (%) (Mean±STD)	Kappa (%) (Mean±STD)
LDA	84.22±2.84	0.83±0.03
Linear SVM	82.50±2.98	0.80±0.04
Quadratic SVM	**86.72±2.52**	**0.84±0.03**
Cubic SVM	86.16±2.63	0.83±0.03
Fine Gaussian SVM	74.68±3.28	0.71±0.05
Medium Gaussian SVM	84.88±2.90	0.82±0.04
Coarse Gaussian SVM	74.72±3.38	0.71±0.05
Fine KNN	80.78±2.84	0.79±0.04
Medium KNN	71.32±3.50	0.69±0.06
Coarse KNN	70.15±3.51	0.68±0.06
Cosine KNN	69.56±3.41	0.67±0.06
Cubic KNN	71.55±3.57	0.69±0.06
Weighted KNN	83.77±2.84	0.80±0.05
Mean	**78.53**	**0.75**

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